

# Module 06: Multiversion Concurrency Control (MVCC) and MVTO

AKTU BCS501 | Unit 5: Concurrency Control Techniques

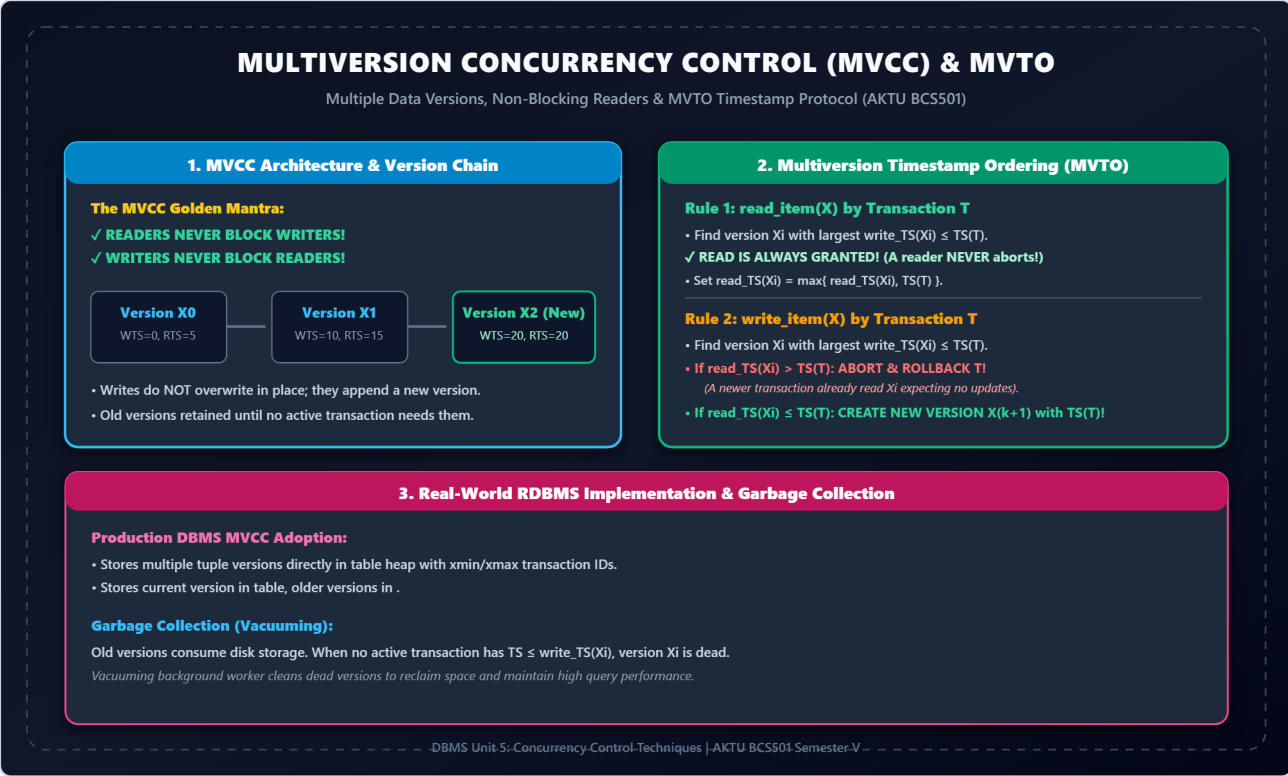


Figure 6.1: Multi-Version Timestamp Ordering Protocol & Version Chain

## 1. MVCC Architecture

In MVCC, writers append new versions rather than overwriting data in-place. Readers access older committed snapshots. Result: **Readers never block writers, and writers never block readers!**

## 2. MVTO Operational Logic

**read(X):** Reads the version  $X_i$  with maximal  $write\_TS(X_i) \leq TS(T)$ . Read is ALWAYS granted! Sets  $read\_TS(X_i) = \max(read\_TS(X_i), TS(T))$ .

**write(X):** Finds  $X_i$  with maximal  $write\_TS(X_i) \leq TS(T)$ . If  $read\_TS(X_i) > TS(T) \rightarrow$  ABORT T. Else creates new version with  $TS(T)$ .